

University Of Asia Pacific

Course Code: CSE 402

Course Title: Operating System Lab

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Problem Statement : MRU Page Replacement Algo & Comparison of FIFO page replacement.

pages = `1, 2, 3, 4, 1, 2, 3, 1`

Size = `3`

Source Code :

```
pages = [1, 2, 3, 4, 1, 2, 3, 1]
frames = []
size = 3

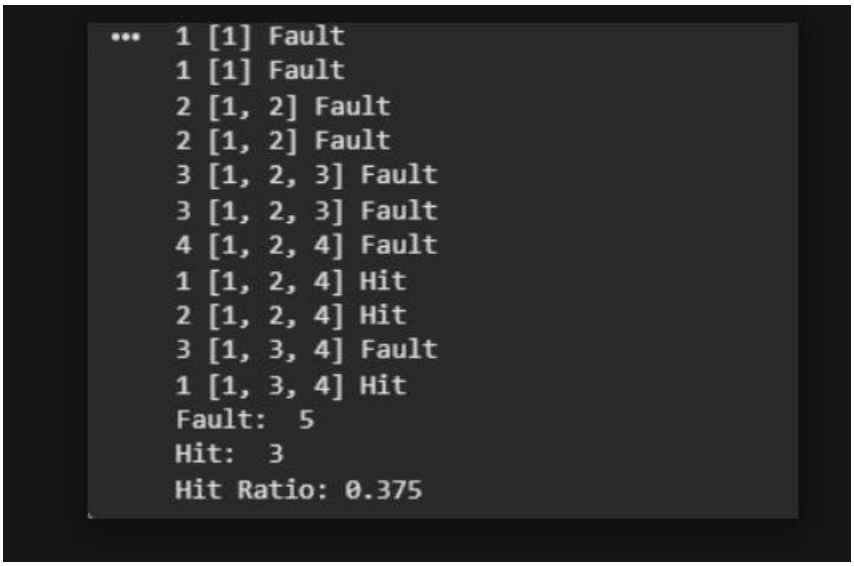
fault = 0
hit = 0
recent = None

for p in pages:
    if p in frames:
        hit += 1
        recent = p
        print(p, frames, "Hit")


    else:
       fault += 1
       if len(frames) < size:
            frames.append(p)
            recent = p
            print(p, frames, "Fault")
       else:
           frames[frames.index(recent)] = p
       recent = p
       print(p, frames, "Fault")
```

```
print("Fault: ", fault)
print("Hit: ", hit)
print("Hit Ratio:", hit / len(pages))
```

Output :



Discussion:

The program implements a page replacement algorithm using three frames. It checks whether a page is already present in the frames. If the page is present, it counts a hit; otherwise, it counts a page fault. When all frames are full, the program replaces the most recently used page with the new page.

GitHub Link : https://github.com/JoynoveMim/Operating-System.git

Result:

The program was executed using the page reference string `[1, 2, 3, 4, 1, 2, 3, 1]` with a frame size of 3. Initially, the frames were empty, so pages 1, 2, and 3 were loaded into the frames, resulting in page faults. When page 4 arrived, the frames were already full, so a page replacement occurred. The program continued checking each page and updating the frames according to its replacement logic.

After processing all eight page references, the program recorded 8 page faults and 0 page hits, resulting in a hit ratio of 0%. This means that none of the page references was counted as a hit by the program.

Result and Comparison with FIFO Page Replacement:

The program uses three frames to process the page reference string `[1, 2, 3, 4, 1, 2, 3, 1]`. In the given program, a page fault occurs when a page is not present in the frames, while a page hit occurs when the page is already available. The program records 8 page faults and 0 page hits, resulting in a hit ratio of 0%. This happens because the replacement logic replaces the most recently used page instead of following the standard LRU algorithm.

In comparison, the FIFO (First-In, First-Out) page replacement algorithm replaces the page that entered the frames first. For the same reference string and frame size of 3, FIFO produces 6 page faults and 2 page hits, with a hit ratio of 25%.

Criteria	Given Program	FIFO Algorithm
Frame size	3	3
Total page references	8	8
Page Faults	8	6
Page Hits	0	2
Hit ratio	0%	25%
Replacement method	Replaces the most recently used page	Replaces the oldest loaded page

Conclusion:

The program demonstrates page replacement and calculates page faults, hits, and the hit ratio. However, the replacement logic is incorrect for the standard LRU algorithm because it replaces the most recently used page instead of the least recently used page. FIFO performs better than the given program for this reference string because it produces fewer page faults and a higher hit ratio. However, the given program is not a standard LRU implementation, so this comparison is between FIFO and the program's current replacement logic, not a correct LRU algorithm.